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From Concept to Practice: A Blockchain-Based Solution for Secure and Efficient Academic Credentials in Higher Education

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ABSTRACT

The generation and verification of academic credentials, particularly diplomas, is crucial for the secure functioning, enhanced legitimacy and confidentiality of higher education institutions (HEIs). Digitising the internal services of HEIs is of great importance, as it mitigates the potential abuse of diplomas and other important documents. The prevalence of forgery attempts highlights the need for robust verification methods. In this regard, blockchain technology can be considered as a solution due to its capability to ensure the immutability, decentralisation, security and transparency of all documents stored on the blockchain network or storage. In this paper, we aim to move from concept to practical implementation of a blockchain-based framework for generating and verifying diplomas, named Diploma Integrity Authentication Record (DIAR), which, in addition to the digitalisation of diplomas and academic documents, also enables their fast, secure, and cost-effective verification. Through the paper, we describe in detail the process of practical implementation of the DIAR framework, presenting the smart contract for generating and verifying diplomas, the process of encrypting and storing diplomas as non-fungible tokens on the blockchain network or storage. At the same time, the challenges of practical implementation are also described, which are predominantly related to the scarcity of blockchain solutions for generating and verifying diplomas in HEIs. Significant attention and emphasis are placed on the legal matters pertaining to the implementation and safeguarding of students' identification and data privacy.

1 | Introduction

The adoption of blockchain technology (BT) in education, particularly in higher education institutions (HEIs), currently faces numerous practical challenges [1], hindering the deployment of blockchain-based systems for managing sensitive information like academic credentials. Despite these obstacles, BT remains the most promising avenue for comprehensively digitalising the creation and verification of academic documents, even though long-term implementations with proven practical effectiveness are yet to be established [2]. In contrast to centralised systems, decentralised blockchain-based solutions offer distributed task

management, enhanced data security, diverse identity options, and robust protection of data privacy and integrity. The inherent characteristics of BT make it currently infeasible to decrypt or exploit data within the blockchain network. BT enables automation of processes through smart contracts, allowing transactions to be executed without intermediaries. An important role in the blockchain network is also played by the distributed ledger, in which every action performed by any member of the network is stored, and the same requires confirmation from everyone to be approved and processed further. This, in fact, represents the transparency and decentralisation of transactions. BT also enables the realisation of various online payments, including

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student payments, even using cryptocurrencies created by the HEI itself, with which all services in that institution will be processed [3]. Even though BT, artificial intelligence (AI), internet of things (IoT) and other contemporary techniques are attempting to professionalise and increase the efficiency and effectiveness of services, including educational services, many challenges still remain, especially issues related to the security, storage and processing of Big Data [4]. Consensus mechanisms play a very important role in creating trust between members of the blockchain network, enabling automation, self-regulating and auditing of smart contracts, as an important part in the functioning of blockchain systems [5]. Convolutional systems currently used in HEIs face various financial challenges [6], centralised diploma management systems [7], slow verification and search of physical files and lack of backup because of the fact that many services are still performed in physical form [8].

While numerous blockchain-based solutions for academic credential management have been proposed and analysed [1, 2], successful long-term implementations in HEIs remain scarce. Our previously introduced framework, Diploma Integrity Authentication Record (DIAR), offers a promising architecture that prioritises security, consensus mechanisms and diploma tokenisation [2]. DIAR presents a pragmatic approach to digitising the generation and verification of academic credentials, particularly diplomas. Building upon this foundation, with the current paper we aim to extend DIAR's functionality to address emerging challenges and opportunities in the field of blockchain-based academic credentialing. Despite the fact that tokenisation is not an innovation in BT, however, to the best of our knowledge, there is no practical blockchain solution that attempts to tokenise diplomas as a whole document without spending additional time and expenses on tokenising diploma credits. The idea of diploma tokenisation comes from the VerDe platform [9], which is a blockchain solution for generating and verifying diplomas by converting earned credits into tokens, and finally, when the required number of tokens is reached, the diploma is generated. However, it is a solution that is not efficient in terms of time and accumulated credits, because it depends on the fixed number of credits generated, which means that if the number of credits changes, then the entire methodology will have to be changed, and diploma verification requires more time and work by verifying all credits. The novelty of DIAR lies not in the improvisation of tokens and blockchain standards, but in their use in the best and easiest form, with the aim of adapting to existing systems, and as cheap as possible the maintenance and cost of creating the system. Meanwhile, the generation and verification of diplomas will be fast and secure.

The current research work makes the following novel contributions:

- An analysis and comparison of several recent blockchain solutions related to diploma generation and verification, and a description of the main advantages of DIAR in terms of time, security, cost and transparency.
- A comprehensive breakdown of the system's role structure and identifying and defining the key actors who can interact with DIAR are defined. Additionally, the mechanisms for

user authentication and the digital signing of diplomas are explored.

- Detailed description of the implementation of the DIAR framework, including user interfaces, with the aim of providing researchers with an overview of smart contracts, progressive web applications, data encryption, and even validation and testing of smart contracts and the system in general.
- A comprehensive analysis and comparison of system testing, utilising specific parameters to evaluate DIAR's performance metrics. This assessment is complemented by a comparative analysis with current diploma management systems employed by HEIs.

The rest of the paper is structured as follows: In the next section, we detail the related work pertaining to blockchain systems applied in education. Next, in Section III, the verification process through blockchain is covered. In Section IV we present the practical implementation of the DIAR prototype, whereas in Section V we conduct a performance evaluation and in Section VI we discuss the findings. Finally, Section VII concludes this paper.

2 | Background and Related Work

The immutability and robust security features of BT enable its application across various domains, including the education sector and HEIs. These features, including decentralisation, make blockchain particularly well-suited for managing sensitive academic data and processes, offering solutions to longstanding challenges in credential generation, verification and record-keeping. Both generation and verification are processes that can be programmed and executed in the form of transactions in the blockchain network, enabling the creation of immutable and easily verifiable transactions [10].

Most of the blockchain solutions that have been created for the generation and verification of academic credentials have used the Ethereum or Hyperledger platform, and recently the Solana network is also being utilised [3, 11–21]. The primary advantages of the Ethereum platform and the reason why we have chosen the Ethereum platform for implementing the DIAR framework include the automation of services via smart contracts, the capability to develop decentralised applications, extensive support for various challenges faced by developers owing to the substantial pool of knowledgeable programmers, the creation of non-fungible tokens representing unique digital assets, and the implementation of consensus mechanisms, among others. Concerns about data privacy and security arise from transparent services, particularly concerns related to data confidentiality stemming from the use of public blockchain. Scalability is a major issue for blockchain-based diploma systems. Diplomas include personal information, making generation and verification difficult, unlike cryptocurrency. The absence of a blockchain approach for successfully maintaining huge numbers of diplomas is a major obstacle for educational institutions considering these systems. This absence of a clear roadmap raises concerns about the system's ability to handle growing volumes of credentials. Moreover, the private nature of diploma data adds layers of complexity to the process. As the number of diplomas

that need to be generated and verified increases, so too might the associated costs, potentially straining resources and limiting widespread adoption. There is also a lack of a blockchain database that has interoperability for high-capacity data management and is compatible with Solidity or other blockchain programming languages [22].

Hyperledger fabric is a promising platform when it comes to the generation and verification of diplomas for several reasons, according to Rani et al. [11]. First, this platform offers privacy protection by offering the possibility that the data is not stored in the public ledger, but the specific participant can access the data in an off-chain form. Second, the owner of the diplomas has full access to the diplomas, selecting who has access to them and what information they can access, providing protection of data privacy. Third, only authorised entities can generate diplomas, thus preventing the fake generation of various fake institutions. And lastly, the data can be easily verified through the hash values of the private data.

Unlike Hyperledger, Ethereum is a blockchain platform that, for the first time, has allowed smart contract programming. It is much easier to use, offering many free development tools with which programmers can test before executing the transaction. Smart contracts have advantages in terms of automating actions without third-party intervention. However, the main drawback of Ethereum is the high cost of transactions, and compared to Hyperledger, it does not allow access permissions [23].

The technologies that are used the most in blockchain platforms for the verification of academic credentials are Web3JS, InterPlanetary File System (IPFS), Progressive Web Application, OpenShift Online and Docker [24]. Speaking about the Ethereum platform, it is also necessary to use the Truffle suite, a framework for the implementation of smart contracts, which consists of Truffle, Ganache and Drizzle, which facilitate the deployment and execution of smart contracts and their more efficient management. Through Ganache, all transactions with all the details of the execution are classified. Metamask wallet enables the execution of transactions through cryptocurrencies, which must first be connected through the private key and the unique address of the user. Among the critical components of blockchain implementation, data storage stands out as a significant challenge. Currently, most solutions rely on either IPFS or centralised databases. This approach is largely due to the absence of a proven, decentralised database solution that has demonstrated effective implementation results in practice. The lack of a robust, decentralised storage solution remains a key area for improvement in blockchain-based systems, particularly for applications requiring extensive data management capabilities [25].

In a blockchain network with multiple nodes, such as the one used for generating and verifying diplomas, numerous users participate in the system. To ensure the integrity of transactions and maintain synchronisation across all copies of data in the distributed ledger, it is essential to establish a key agreement among all nodes. One of the advantages of distributed ledgers is that they are resistant to malicious behaviour by a small set of nodes, giving them a highly accessible and decentralised digital infrastructure. However, DLT (Distributed Ledger Tech-

nology) has several downsides regarding scalability and privacy. Blockchains are a subset of DLT and are probably the most widely used [26]. The main feature of blockchain is that transactions are aggregated into blocks and each block of data carries the hash value of the previous block. In Figure 1 the blockchain process is shown (i.e., transaction processing), which in our system is the generation of the diploma or its verification.

The current state of the art reveals a proliferation of blockchain-based prototypes designed for diploma generation and verification. These innovative solutions demonstrate the growing interest in leveraging blockchain technology to enhance the security and efficiency of academic credential management. Although numerous conceptual models and experimental prototypes have been proposed, including numerous practical solutions, there is no research that shows that any blockchain system is being successfully deployed and used in any HEI. UAE-chain is a prototype based on a blockchain framework for validation, authentication and equivalence of academic credentials [3]. This prototype gives special importance to secure authentication through BT, where the certificate in a transparent form can be signed using Elliptic Curve Digital Signature Algorithm (ECDSA). A prototype that also uses ECDSA for digital signature and verification is Educert-chain. It is a platform that gives special importance to the process of safe notarisation of certificates using the permissioned blockchain, hyperledger fabric platform, the SHA-256 algorithm for cryptography mechanisms and CouchDB as a static database for data storage [11]. VECefblock is another prototype that is deployed on the Amazon EC2 cloud [12]. It has a defined architecture, several performance experiments, and is among the first research projects that show the use of blockchain technology in Vietnam. Decentralised verification of diplomas (DVDs) is another blockchain solution that aims to verify certificates using Self Sovereign Identity (SSI), respectively, standards such as decentralised identifier (DID) and verifiable credential (VC) [13]. Next, the BeCertify platform presents a real prototype, which has been tested and validated at IEBS Business School [15]. It supports the use of disruptive BT and aims to digitise certificates that are generated in the traditional form. The digital token pays attention to the tokenisation of diplomas, respectively, the conversion of non-fungible values into unique tokens.

A slightly different prototype is Creducate, which was developed in the Solana network due to the faster processing of transactions and the lower costs of executing transactions [16]. QualiChain is considered by the authors as the first ecosystem for the verification of certificates on a wider scale [17]. The prototype uses cryptographic mechanisms such as SHA-2 and SHA-3, and IPFS for certificate storage. Another case is IU-TransCert, which uses a data structure called Auditable Merkle Tree that enables the generation of credentials and other auditing capabilities [18]. Credere is another platform for processing, storing and sharing digital academic credentials. It is a system that uses ES-256-CBC encryption as a security mechanism to communicate with APIs [19]. B4E is a system for creating and verifying academic credentials, using BT and remote signing. B4E works in cooperation with other external systems, such as Bsign and the BNB network [20].

Lastly, DIAR is a comprehensive blockchain-based framework developed on the Ethereum platform [2]. It employs a

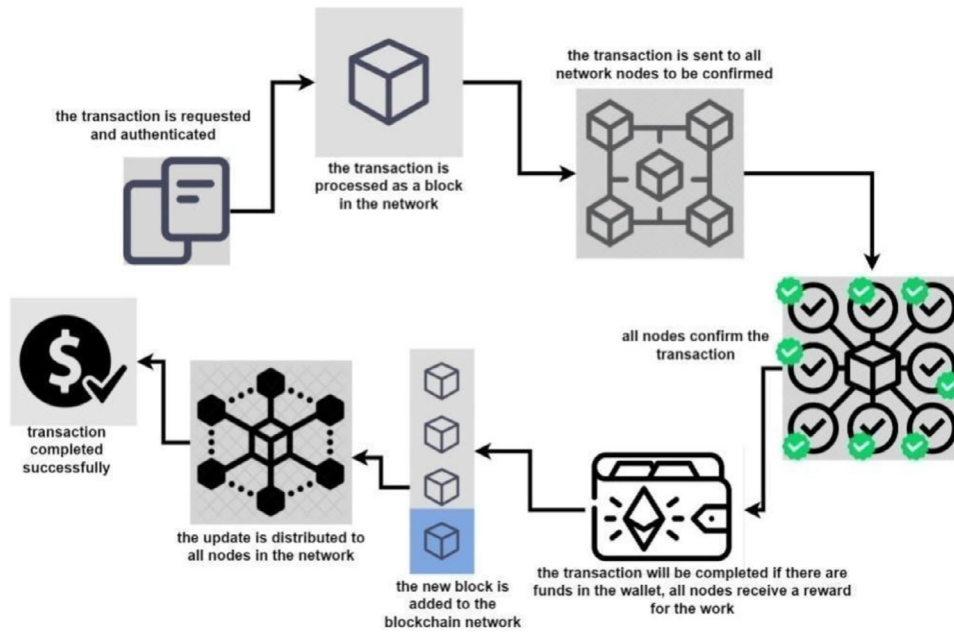


FIGURE 1 | The blockchain process for realising a transaction (diploma).

sophisticated multi-layered network architecture that addresses key aspects of security, consensus mechanisms and efficient data transfer. The system features a well-defined structure, including a detailed conceptual model and systematic architecture, which clearly outline the processes for generating and verifying academic credentials.

In Table 1, we showcase a summary of the features of the most recent blockchain solutions for the generation and verification of academic credentials, including the proposed DIAR framework. The table encompasses critical aspects such as practical implementation, blockchain platform, security, efficiency, compliance and storage type, providing a clear perspective on the strengths and innovations of each approach. This paper aims to further develop and refine the DIAR framework, building upon existing blockchain solutions in the field of academic credential management, which are not included in our previous research [1, 2].

In terms of data security, as specified in the architecture defined in [2], DIAR aims to combine at least three techniques, combining AES for data encryption, SHA-256 for verifying the identity of system users and any of these algorithms, such as Zero-Knowledge Proofs or Homomorphic Encryption, regarding data and user privacy issues. The efficiency of the system is based on the fact that it does not perform tokenization of credits or generate more transactions, which are undoubtedly also costly when talking about blockchain networks. Instead, DIAR stores the diploma as a document in the form of a token, with a unique address and hash value, which guarantees the immutability of the data within this document. In the event of verification, very quickly and easily, the student's data, respectively the diploma, can be verified as original or not. Regarding compliance, DIAR, as a blockchain-based system, guarantees the immutability, transparency and automation of diplomas through its design and functionality [27]. This is achieved through the use of smart contracts, the Ethereum platform and security mechanisms that protect data,

ensure secure authentication and preserve the privacy of all users in the system.

This paper also clarifies the execution of transactions, offering developers a clearer understanding of how to practically implement a blockchain prototype and conduct transactions on the Ethereum platform using the Sepolia test network. Many blockchain solutions lack detailed information or elaborate explanations of the practical implementation of these processes. This research seeks to fill this gap by providing a comprehensive, step-by-step explanation of the DIAR framework, illustrated through graphical representations, pseudocode and performance metrics analysis. Our objective is to provide a fundamental resource for novice and intermediate researchers and developers in blockchain programming, facilitating their understanding and development of blockchain systems. Figure 2 highlights the procedure for encrypting and decrypting diplomas, hence, explaining these ideas.

3 | Verification Through Blockchain Systems

By streamlining the verification process, blockchain not only makes things more efficient but also helps build greater trust in the educational system as a whole. Figure 3 illustrates how this transaction process works within the blockchain network in its simplest form. Educational institutions can generate diplomas by storing them as tamper-proof records in the blockchain network, as well as verify credentials without any central authority [28]. Verification in blockchain systems involves a critical process that consists of four key stages, as outlined in [29]. First, the unique hash value of the diploma must be verified to ensure it has not been altered. This involves checking all details associated with the diploma. Second, the date of issue needs to be confirmed. The actual date of generation is recorded digitally at the time the diploma is created, and it must be compared with the verification date to ensure that the verification occurs after the diploma's

TABLE 1 | Summary of related blockchain solution.

Study	Name	Practically— tested	Blockchain platform	Security	Efficiency	Storage	Compliance
[3]	UAEchain	Yes	Ethereum	ECDSA algorithm, elliptic curve cryptography (ECC)	Yes	Public key cryptography (PKC) and MongoDB as state database	Yes
[11]	Educert-chain	Yes	Hyperledger fabric	Elliptic Curve Cryptography (ECC), SHA-256	Yes	CouchDB as state database, Docker container for portable workloads	No
[12]	VECefblock	Yes	Hyperledger fabric	*not specified	No	IPFS (InterPlanetary File System)	No
[13]	DVD	Yes	Hyperledger Indy Ledger	Self-sovereign identity (Decentralised Identifier [DID]) and Verifiable Credential (VC)	No	Hyperledger Aries Cloud Agent Python (ACA-Py) library	No
[14]	BeCertify	Yes	Ethereum or polygon	Europe Chain	No	IPFS (InterPlanetary File System)	No
[15]	Digital Token	Yes	Python 3.10.8	NFT, proof of work	No	NFT token, storage not specified	No
[16]	Creducate	Yes	Solana	Anchor framework, API calls	No	online storage (dropbox)	No
[17]	QualiChain	Yes	Ethereum	Blockcerts Wallet, Public-Key Infrastructure, Certificate Authorities (CAs)	Yes	IPFS, SWARM or Metadisk	Yes
[18]	IU-TransCert	Yes	Ethereum	Auditable Merkle Tree structure, STRIDE threat model	Yes	On-chain storage, smart contract	Yes
[19]	Credere	Yes	Hyperledger	AES-256-CBC	Yes	distributed-ledger technology for on-chain, MySQL for off-chain	Yes
[20]	B4E	Yes	Hyperledger FireFly	softHSM, BSign Architecture	Yes	Not specified*	Yes
[21]	DocStone	Yes	Rinkeby, ethereum	Transport Layer Security. Architecture Trade-off Analysis Method (ATAM)	Yes	Not specified*	Yes
[2]	DIAR	Yes	Ethereum	AES, SHA-256 and ZKP or HE	Yes	InterPlanetary File System (IPFS), smart contract	Yes

issuance (i.e., the current date must be greater than the date of generation). Third, a check against the list of revoked diplomas is necessary. This list can only be updated by the diploma issuer for various reasons. During verification, it is essential to confirm that the diploma is not on this list. Finally, the authenticity of the student must be validated. This involves ensuring that the student's ID matches the ID generated by the institution that issued the diploma. The accuracy of this matching relies on IDs recorded in the blockchain network during the initial transaction for generating the diploma.

A critical step during verification is obviously finding and verifying the unique hash value stored in blockchain storage, the decryption mechanisms that the system uses for the purpose

of decrypting the data stored as blockchain transactions in blockchain storage (AES or SHA256), and finally the authentication of the student in order for the data to appear in the system. If the correct authentication is not performed, the data will not appear in the blockchain system. We present this process through Figure 4.

4 | Toward Practical Implementation of the Diar Blockchain Prototype

In terms of operation, DIAR is very similar to the VerDe platform [9], a platform that is also based on the tokenisation of student credits, converting the earned credits into fungible tokens, and

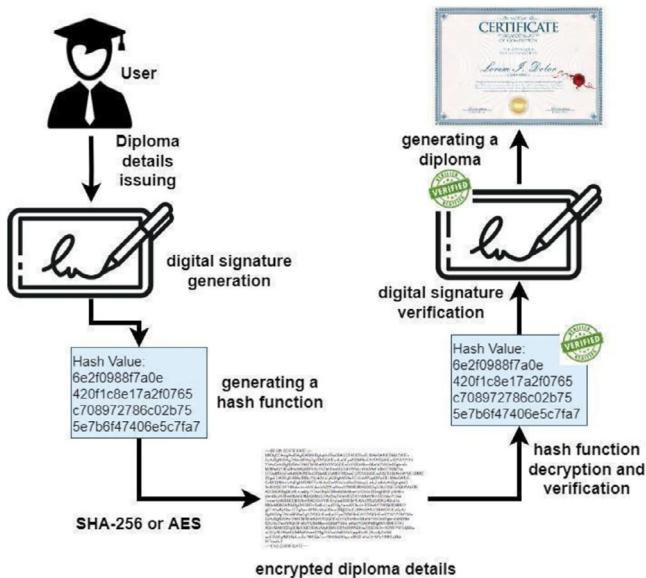


FIGURE 2 | Description of the encryption and decryption process of the diploma.

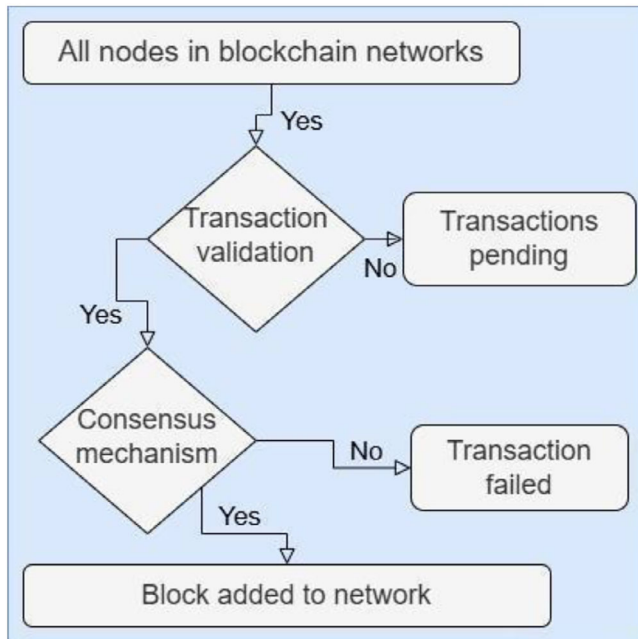


FIGURE 3 | Transaction execution on blockchain network.

storing them in the blockchain network. However, unlike VerDe [9], DIAR aims to keep the complete diploma as a whole document, while access to the diploma is based on the private key and unique hash value that it possesses. Only if it is confirmed that the private key and hash value are identical to what is required can the same be verified and the details within the diploma be accessed. Thus, the efficiency of DIAR is based on the fact that it does not perform tokenisation of credits or generate more transactions, making it a simpler, faster and cheaper solution compared to credit-based tokenisation.

When a diploma is stored in IPFS, a Content Identifier (CID) is generated—a unique label that doesn't reveal where the file is

stored but acts as a decentralised address, ensuring immutability and preventing duplication. Any content change produces a new CID, while identical data across nodes yields the same one, as it depends on the hash value. The CID and related data are stored on the blockchain via smart contracts. During verification, the verifier uses the student's unique ID to retrieve the stored CID, which is then used to locate the diploma in IPFS. Once confirmed as unique and immutable, the verified diploma is displayed, ensuring authenticity and integrity. IPFS is not the only option for blockchain-based storage in systems that generate and verify diplomas. However, the DIAR system uses IPFS because it offers better compatibility for handling larger files, such as diploma PDFs, and is more cost-efficient. Only minimal data, such as the CID and hash value, is stored on-chain, while all other metadata is kept off-chain. This setup provides a more suitable and scalable solution for managing and verifying digital diplomas.

The functional and non-functional requirements of the DIAR system are specified in [2, 30] along with all the use case diagrams and the primary actors that are part of the system, including the use case scenario. In summary, the sequential diagram in Figure 5 illustrates the process of generating or verifying diplomas. The system gives users full control over their personal data. Users can securely manage their identity data, including personal data, credentials and verifications, on the blockchain. This decentralised approach eliminates the need for central identity authorities, ensuring that users are the sole owners of their digital identity.

Regarding the system organisation and defined roles in the system, there are 4 roles in total, including *superadmin*, *admin*, *moderator* and *application programming interface (API)*. The *superadmin* is the highest level of system organisation, which can even delete the entire system organisation or add or set certain permissions for the system user. Then the *admin* has the right to update the details of the organisation already created by the *superadmin* to update the roles of the members of the organisation. In an HEI, such a role can be played by the management of the HEI itself. The *moderator* has the right to add or even delete members of the organisation created by the *superadmin*. The moderator's permissions are strictly limited to the management of members within a specific department or faculty unit. This may include monitoring participation, verifying relevant documents, inviting new members, and issuing notifications regarding updates such as diploma revocations. All privileges associated with the moderator role are granted and governed by the super admin and are constrained to the scope of the designated department or faculty to which the moderator is assigned. And the last level of organisation is the ability to connect this organisation to the system *API*. That is, students, professors, and other academic staff participate here, through which they can create, read and delete data for their personal account. Figure 6 provides a description of the main roles of the system, the organisation of the system and the main actors in HEI.

4.1 | The Generation Process of the Diploma Through Diar

To implement a functional blockchain system, it is important to first define the consensus mechanisms responsible for validating transactions and maintaining network integrity. DIAR uses

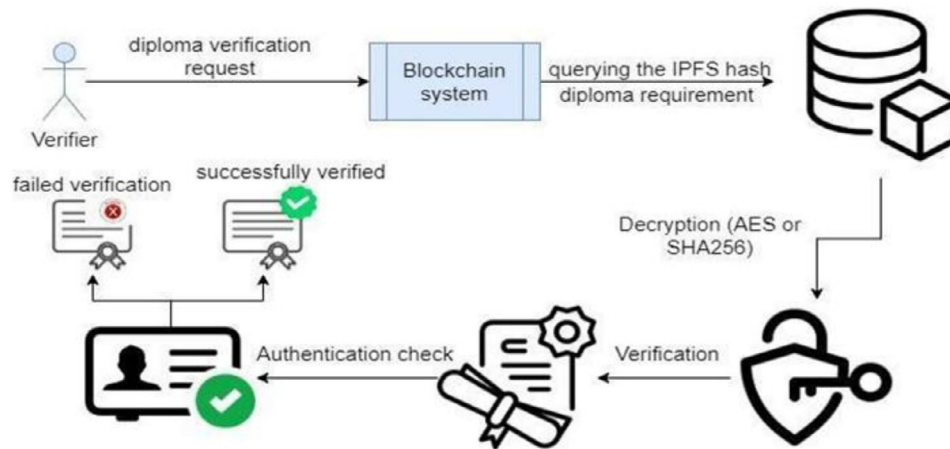


FIGURE 4 | Description of the diploma verification process.

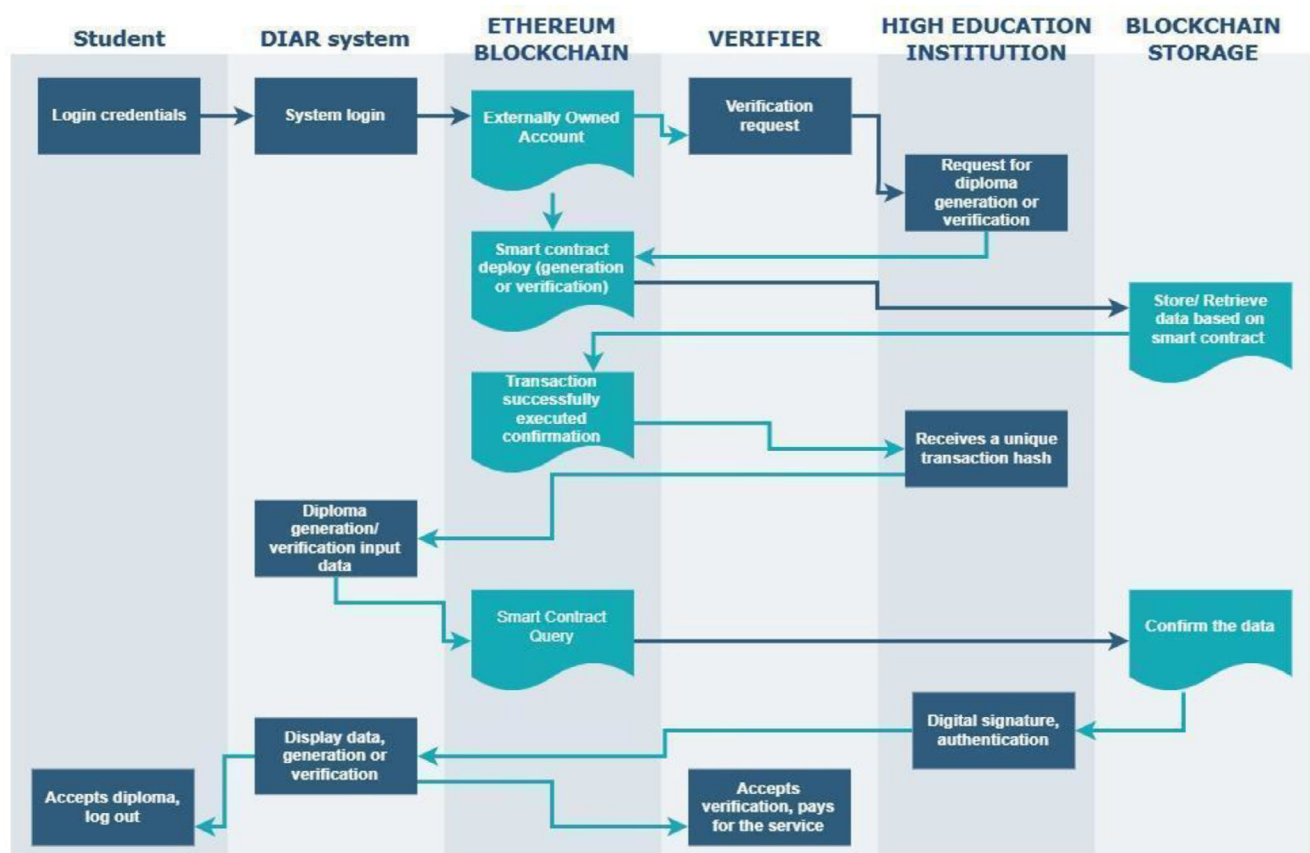


FIGURE 5 | Generation or verification process through DIAR blockchain system.

both proof-of-work (PoW) and proof-of-stake (PoS), leveraging Ethereum's validation protocols. This approach not only simplifies deployment and improves energy efficiency but also supports the decentralisation of activities. These validation processes ensure secure and reliable transaction approval, upholding the integrity of academic credential management. For the current phase of implementation and testing, DIAR relies on mechanisms provided by the Ethereum platform to ensure security, transparency and privacy. In the future, the development of custom validation rules, tailored to the roles and requirements of all

system participants and adapted to the practical needs of higher education institutions, is considered a promising direction.

Regarding the frontend and backend, as well as the communication with blockchain development, Python and Node.js are used. The design part is created using JavaScript, respectively Node.js, HTML and CSS as the basic part for design, React.js as a framework and Web3.js for integrating the system with the blockchain network. Metamask is used for authentication and confirmation of transactions, whereas IPFS has been used for the database as

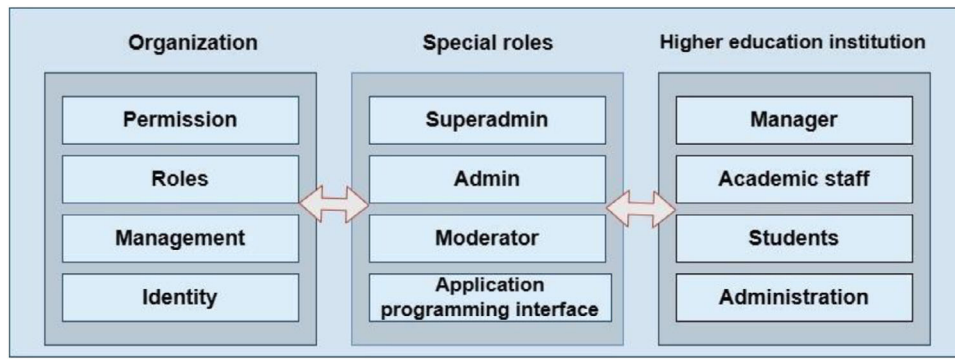


FIGURE 6 | Organisation of roles on the DIAR system.

well as the network's own distributed ledger for testing smart contracts. The implementation of smart contracts was done in Solidity, their testing in RemixIDE and their deployment on the *Ethereum Sepolia* test network.

The system operates as follows: an administrator logs into the DIAR platform through the university's secure portal to process student credentials. With just a few clicks, they upload diploma information, review automated validation checks and issue a new digital diploma, a process managed by DIAR's intuitive dashboard that also allows easy revocation or verification of existing records as needed. A student receives a notification in their LMS profile that their diploma has been issued. They can instantly access their verifiable credential, download it for future use, or share it securely with employers and institutions. The DIAR interface ensures students always have easy, tamper-proof access to their academic records, enhancing trust and usability at every step.

After verifying the information, the student can digitally sign the diploma using special tools that mimic a physical signature. Once they are satisfied, they can click the submit button. This action triggers a confirmation email to the student. After the student confirms their submission and it is signed by management, the diploma undergoes an encryption process before being deployed to the blockchain. This procedure ensures that it is stored with a unique hash value, providing immutability, transparency and the ability for online self-identification and digital signature verification. Figure 7 illustrates the interface used for signing the diploma before it is stored as a token in the blockchain database.

Any future changes to a diploma stored in the blockchain database, whether made by HEI management or any other authorised party, will result in the generation of a new hash value. The system promptly notifies all involved parties through an activity log and sends an email alert. As a result, the document will no longer retain its original hash value; instead, each modification creates a unique hash that differs from the initial one. This mechanism provides a clear indication of whether the diploma has been altered, making it easy to verify its authenticity in the future by comparing it to the original hash value established at the time of entry. Figure 8 illustrates the process of inserting the diploma as a transaction in the blockchain network.

The process of connecting to the Ethereum network, creating instances of smart contracts, interacting with contracts and the

functions of inserting, generating, and retrieving data have been explained in detail through Algorithms 1, 2, 3 and 4, presented as pseudocode in a subsequent manner. The complete source code for the implementation, including also Algorithms 5 and 6 are publicly available in GitHub. This includes the Solidity and JavaScript code used in the project. The GitHub repository can be accessed via the following link: https://github.com/avnirustemi2025/BLOCKCHAIN-SYSTEM_DIAR

Algorithm 1 shows the establishment of the connection with Ethereum Mainnet. Initially, we check for the availability of Ethereum providers, and subsequently, a request for an account is sent. If this is successful, and the networkID is valid, connection and interaction with the smart contract will occur. On the contrary, for every error and disconnection, it will give the appropriate message, indicating where the problem lies.

Algorithm 2 shows the initialisation of the smart contract instance, namely the initialisation of the contract address and ABI, so that interaction with the code in Solidity can then be done, depending on which smart contract is executed. It is very important to initialise the ABI values and the address of the smart contract, because initialisation and calling of functions within them is only done if both of these are correct. Only after the connection with the account and provider is made, can deployment of the smart contract begin, wherein this specific case we have the insertion of diploma data and the retrieval of data to check that the data has been saved.

Algorithm 3 illustrates the process of inserting and retrieving the stored data. To achieve this, we must ensure that steps of Algorithm 1 and 2 are successful, namely connecting to the blockchain node and initialising the address and ABI of the contract. Then, the interaction with the smart contract coded in Solidity comes into focus, where the functions that are part of the contract are called for inserting, generating or even retrieving data from blockchain storage.

With Algorithm 4, we show the smart contract for generating the diploma. Initially, as with the procedure in Algorithm 3 where we had data insertion, the steps of Algorithm 1 and 2 must be successful. Thus, the connection to the Metamask provider, as well as the connection to the ABI and the contract address, must be successful. In this algorithm we have 2 smart contracts, for issuing the diploma and verification. Initially, the connection to the student's respective account is required, and then the

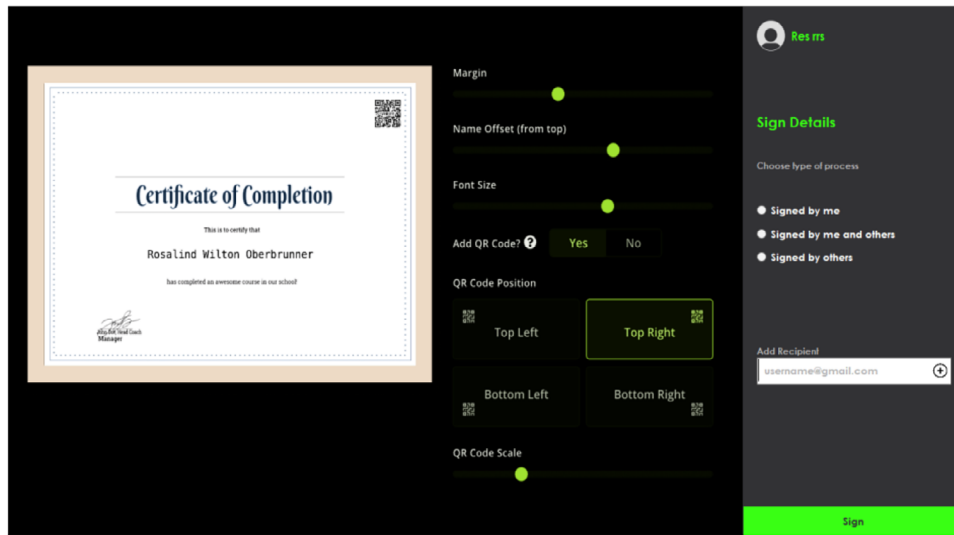


FIGURE 7 | Example of signing the diploma during insertion.

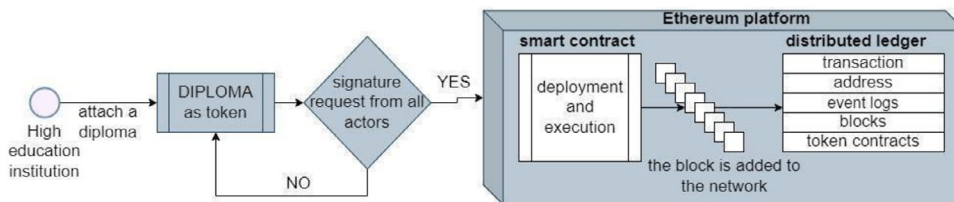


FIGURE 8 | Issue process of the DIAR system.

execution of the *IssueDiploma* function begins where, among other things, the values are associated with each parameter that is inserted by the system user, and then the same data is given a unique hash value, and an attempt is made to generate the diploma along with the confirmation number, otherwise, it gives a message for any errors. The verification function is used to verify that the diploma has been issued and that everything is in order with the generation of the relevant diploma, which obviously depends on the inputs provided by the relevant student.

The definition of constructors is important in Solidity, because of their need during the deployment process. They are used to initialise variables that are required before the contract becomes operational. The definition of structures is similar to that of other programming languages: to organise data, reduce costs and decentralise data. Mapping is used to search or verify data we already have in the blockchain storage, which has an identification value, a hash value, or an identification generated during insertion. Function *IssueDiploma()* – is used to generate a diploma based on the data we provide, and we map them to the defined certificates, but if they do not exist, on the contrary, the transaction will not be performed. Function *RetrieveDiplomaData()* – allows receiving data already inserted into IPFS or another blockchain storage and extracting it to the user's front-end or web form. Generating the hash value of the document, or rather the diploma, which will be stored as an immutable token, can be done using SHA256 if it is an off-chain blockchain, while we are on-chain, the same can be stored in a smart contract and verified whenever we need it.

4.2 | Verification of Diplomas Through the Blockchain System

Verification of diplomas is an easier process if their insertion is done correctly, without errors. In the system, the super admin has the right to add or remove *contractors* and check diplomas, which obviously must be in agreement with the HEI management.

Through Algorithm 5, we have introduced the verification of diplomas in the DIAR blockchain system. To verify the diploma, first the connection must be made with the account and the provider. Then the process of verifying the unique hash value of the respective diploma, the student's address and ID begins, where the verification is mainly based on these last two elements, namely the ID and confirmation number of the diploma generation. If these two elements are correct, which means that such a degree exists in the blockchain storage and as such has not undergone any change (i.e., has the same unique hash value), then the verification will be successful.

For the verification of diplomas in a blockchain system, including the DIAR system, several important processes must be taken into account, such as the deployment of the smart contract depending on the given inputs and the prior agreement between the verifier and the HEI; the IPFS hash return, which depends on the address of the corresponding student's diploma; verification which confirms where is it compared the received diploma with the information provided by the recipient; authenticity check which has to do with the verification and authentication of the

INPUT: Expected Network ID
OUTPUT: Connection Status (Connected or Error)

1. **IF** an Ethereum provider is available **THEN**
2. **LOG** “Ethereum provider found”
3. **REQUEST** user account access
4. **IF** account access is granted **THEN**
5. **GET** current Network ID
6. **LOG** “Current Network ID: [Current Network ID]”
7. **IF** Current Network ID is equal to Expected Network ID **THEN**
8. **LOG** “Connecting to Ethereum Network”
9. **INITIALIZE** Web3.js connection
10. **ATTEMPT** to get a block number from the network
11. **IF** successful **THEN**
12. **LOG** “Connected to Ethereum Network”
13. // Proceed with smart contract interaction (implied)
14. **ELSE**
15. **LOG** “Error getting block number”
16. **ENDIF**
17. **ELSE**
18. **LOG ERROR** “Wrong network, please connect to the expected network”
19. **ENDIF**
20. **ELSE**
21. **LOG ERROR** “Error requesting accounts”
22. **ENDIF**
23. **ELSE**
24. **LOG ERROR** “Ethereum provider not found. Please install a provider.”
25. **ENDIF**

diploma data in coordination with the HEI management. And finally, if the diploma meets the expected criteria and is verified as authentic, the verification process is successful. Otherwise, further investigation may be required.

5 | Performance Evaluation

The prototype we developed facilitates the diploma tokenisation process, significantly simplifying how diplomas are managed. Each diploma, saved in PDF format, is digitally signed by all necessary actors—the student, the rector and the dean—before being stored as a non-fungible token in the database. This database is secured with advanced encryption algorithms like AES and SHA256, ensuring that the diplomas are protected. Importantly, this system allows for easy confirmation of the diplomas at any time, enhancing both accessibility and security.

INPUT: Contract Application Binary Interface (ABI), Contract Address
OUTPUT: Contract Instance (Initialized or Error)

1. **SET** Contract ABI to provided value
2. **SET** Contract Address to provided value
3. **IF** an Ethereum provider is **NOT** available **THEN**
4. **SET** Web3 instance to the browser’s Ethereum provider
5. **ATTEMPT** to request user account access
6. **IF** account access is granted **THEN**
7. **INITIALIZE** Contract Instance using Contract ABI and Contract Address
8. // Proceed to interact with the smart contract
9. // (e.g., call functions like GenerationDiploma, VerificationDiploma, RetrieveData)
10. **ELSE**
11. **LOG ERROR** “Error requesting accounts”
12. **ENDIF**
13. **ELSE**
14. **LOG ERROR** “Ethereum provider not found. Please install a provider.”
15. **ENDIF**

INPUT: Contract ABI, Contract Address, Input Data (for insertion), Identifier (for retrieval)
OUTPUT: Insertion Status, Retrieved Data (or Error)

1. **SET** Contract ABI, Contract Address
2. **IF** Ethereum provider is **NOT** available **THEN ERROR** “Provider not found”
3. **ATTEMPT** Request Account Access
4. **IF** Access Granted **THEN**
5. **INITIALIZE** Contract Instance
6. **FUNCTION** InsertDiplomaData(Data):
7. **ATTEMPT** Get Accounts
8. **IF** Accounts Available **THEN ATTEMPT** Call ‘insertData(Data)’ on Contract
9. **ELSE ERROR** “No accounts”
10. **END FUNCTION**
11. **FUNCTION** RetrieveDiplomaData(ID):
12. **ATTEMPT** Call ‘retrieveData(ID)’ on Contract, Store Result
13. **END FUNCTION**
14. **ELSE ERROR** “Account access denied”
15. **ENDIF**

INPUT: Contract ABI, Contract Address, Diploma Data, Student ID

OUTPUT: Issue Status (Hash), Verification Result (or Error)

1. **SET** ABI, Address
2. **IF** No Ethereum Provider **THEN ERROR** “No Provider”
3. **ATTEMPT** Request Accounts **ELSE ERROR** “No Account Access”
4. **INITIALIZE** Contract
5. **FUNCTION** IssueDiploma(Data):
6. **ATTEMPT** Get Accounts **THEN ATTEMPT** Call ‘IssueDiploma(Data)’ -> Hash
7. **ELSE ERROR** “No Accounts”
8. **RETURN** “Issued”, Hash (or Error)
9. **END FUNCTION**
10. **FUNCTION** VerifyDiploma(ID):
11. **ATTEMPT** Call ‘VerifyDiploma(ID)’ -> Result **ELSE ERROR** “Verification Failed”
12. **RETURN** “Verified”, Result (or Error)
13. **END FUNCTION**

INPUT: Diploma File, Student Address

OUTPUT: Verification Status (True/False)

1. **SET** Contract ABI, Contract Address, IPFS Node URL
2. **IF** No Ethereum Provider **THEN ERROR** “No Provider”
3. **ATTEMPT** Request Accounts **ELSE ERROR** “No Account Access”
4. **INITIALIZE** Contract, IPFS Client
5. **FUNCTION** UploadAndVerify():
6. **GET** Diploma File
7. **ATTEMPT** Upload Diploma to IPFS -> Diploma Hash **ELSE ERROR** “IPFS Upload Failed”
8. **ATTEMPT** Call ‘AddDiplomaHash(User Address, Diploma Hash)’ on Contract
9. **ATTEMPT** Call ‘VerifyDiploma(Student Address, Input Diploma Identifier)’ on Contract -> Verification Result
10. **RETURN** Verification Result
11. **END FUNCTION**

Initially, the system offers digital identification methods, which the user himself determines. Diplomas are digitally signed, with the student confirming the identity after signing, and then the document is again digitally signed by the institution itself and encrypted. In this way, the diploma already has unique values, and if the user tries to change something, such a change is very difficult, that is, impossible.

To initiate any changes in the blockchain network, approval must be obtained from all nodes, as higher education institutions involve multiple stakeholders. Additionally, it is important to note that token data cannot be updated once it is stored; instead, any modification will generate a new hash value. This new hash serves as a reliable indicator of the diploma’s authenticity. Moreover, it is essential to ensure that the system adheres to relevant regulations and standards related to data protection and educational accreditation.

There are many web-based integrated development environments specifically designed for Ethereum smart contract development. RemixIDE, as one of them, offers a suite of tools and features to streamline the process of writing, testing, debugging and deploying smart contracts on the Ethereum blockchain. Every smart contract, before it can be deployed, whether in RemixIDE or any application that enables its execution, must have our unique address associated with the Metamask wallet address, which, before it can be deployed, requires verification by the tools we have in Metamask. Only after this is confirmed can it be executed, but each execution has its own cost. If we analyse Figure 9, on the right-hand side where we have the Metamask wallet, we also have the estimated transaction execution fee, which shows us the cost of executing that smart contract.

Many applications that allow smart contract programming, compilation, and debugging also include a smart contract testing library, which both shows errors and provides additional clarification. Again, we take RemixIDE as an example, since it also provides free services. In Figure 10, it is clear that it allows testing of the smart contract but also includes the AI agent, which helps the developers where needed, giving additional explanations for whatever they execute. The help offered by AI robots is more varied, but among the most important are code generation, static analysis and security, code documentation and help, custom plugins and more.

As for the smart contracts created, they are deployed using the SepoliaTestnet, which is essentially a testnet used by developers to experiment with new blockchain technologies or applications without risking real assets. One of the main features of the blockchain is the transparency of the smart contract. Figure 11 shows how the smart contract can be tested directly in Etherscan, as with web services, we can test their execution; however, here it immediately asks for confirmation of the execution of the smart contract in Metamask. Which means that, in the Ethereum blockchain network, every service must be paid for with Ether.

Figure 12 shows the successful verification of the diploma, based on the unique index of the student, whether it is stored or not in the blockchain network, and gives us the unique number of the degree. Then we tested whether that diploma with a unique number exists in the blockchain database or not, and from the figure we see that it is valid, and it does exist. This is actually just a short string from the verification smart contract, which truly shows the successful validation of the smart contract and that they are functional.

When talking about metrics and system performance, it is difficult to determine and measure exactly which framework is better than the other, since many indicators must be taken

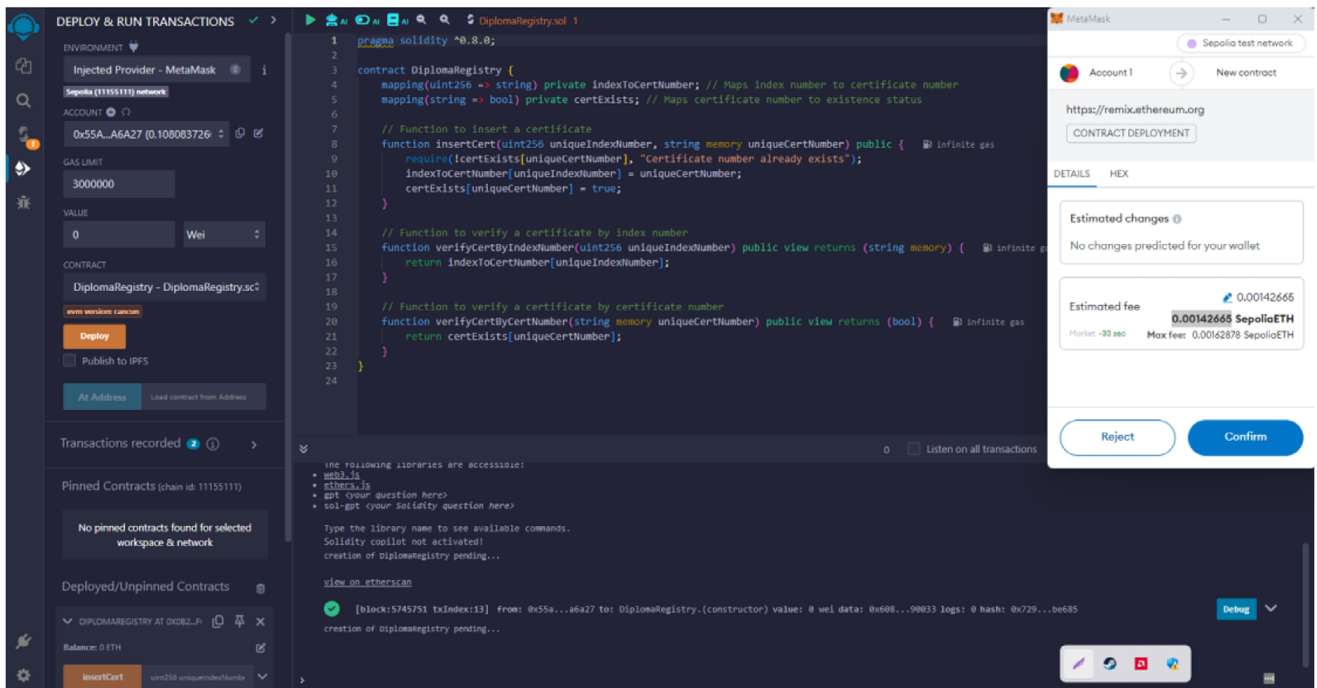


FIGURE 9 | Estimated transaction fee, pending confirmation from Metamask wallet.

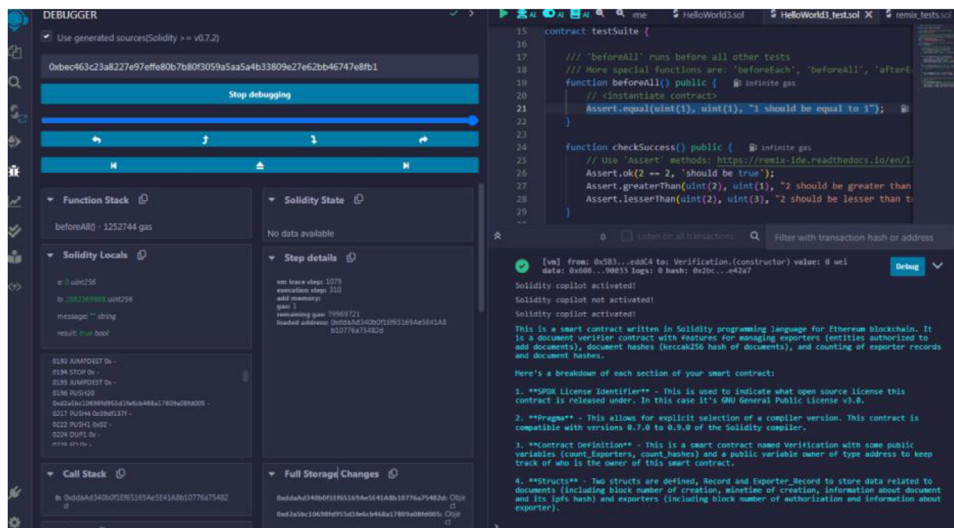


FIGURE 10 | The process of testing smart contracts and including intelligent agents in RemixIDE.

into account in this regard. There are even different types of metrics that we can analyse, depending on whether we analyse performance, security, cost, usability, etc. For us in this regard, we have analysed the performance of the system, mainly to calculate the performance metrics, as the most important indicators are transaction throughput capacity, transaction latency, production network, testing network and transaction cost [31]. Algorithm 6 outlines how to measure the performance (especially gas usage) of smart contract functions using a testing and deployment environment like Hardhat. Very important for the performance of the transaction execution is the balance and gas limit, which we determine during the execution of the contract. The more we increase the gas fee, the faster the same transaction can be executed. Through the algorithm, we have presented some main

elements that must be taken into account, namely the balance we have in the portfolio or the respective provider, the start gas that we determine, which depends on the gas fee of the respective contract, since today, thanks to advanced IDEs, we can see in advance how much fee we need for execution, and finally the new balance is calculated, which is in direct proportion to the total gas fee that has been spent during the execution of the smart contract.

All transactions in Ethereum are carried out through accounts, which participants must have. There are two types of accounts: externally owned accounts (EOAs) and contract accounts (CAs). Both are interconnected, the difference being that CA is a deployment transaction while EOA stores the entire history of transactions that the user has made [32]. Not all TPS metrics are

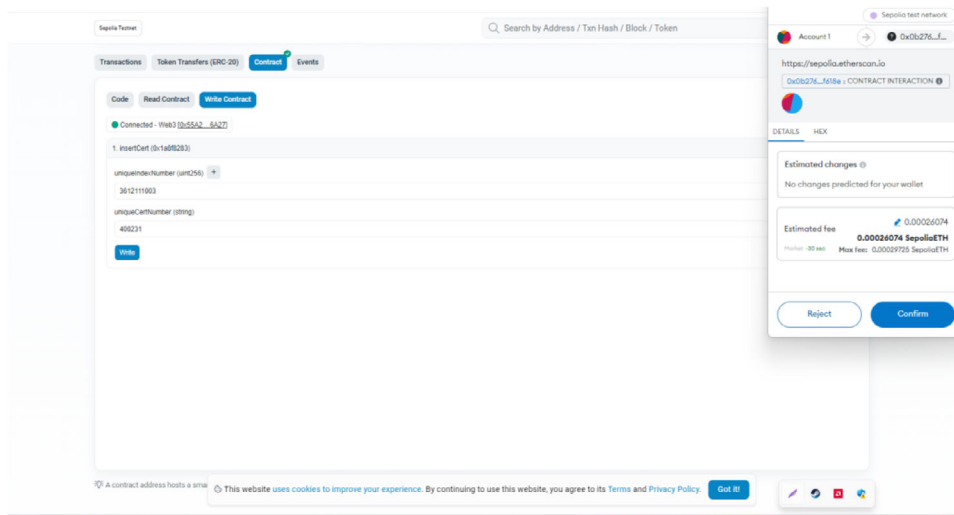


FIGURE 11 | Testing the smart contract in Etherscan.

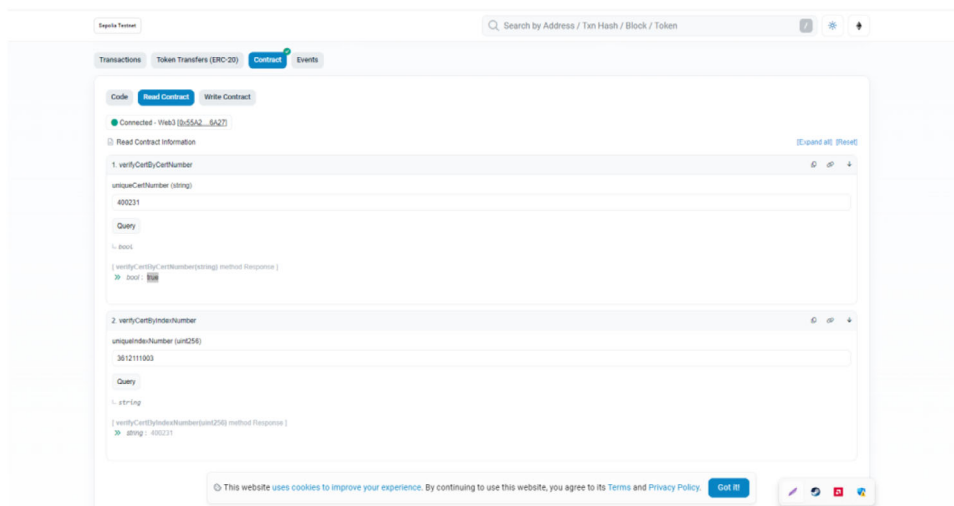


FIGURE 12 | Diploma verification based on student ID and diploma number.

the same. There are 3 main types of metrics that are related to the real-time execution of transactions, the maximum number of transactions that the blockchain network has processed, and the maximum number of transactions that it can theoretically process based on its parameters. Undoubtedly, one of the most difficult is real-time TPS, because certain time intervals, hours, days, weeks, or even months, are needed to calculate the number of transactions in that time interval and divide the same by seconds (depending on the time interval) [33].

A platform similar to Ethereum, which is also showing great interest from many participants due to its public nature, but at the same time higher transaction processing speed and higher costs, is Solana, which uses its own token called Sol. Solana processes 3,000% more transactions than Ethereum, has around 1300% more daily active users, and its transaction fees are nearly 5 million percent cheaper. This is one of the main advantages that this platform offers, and it has made users active on this platform, even though it is new to the blockchain network market. However, there are many shortcomings in terms of the devel-

opment environment, in terms of smart contract development, their transparency, and development tools that enable testing, development and execution of smart contracts [34]. Table 2 compares the main platforms used in the creation of blockchain systems for generating and verifying academic credentials, as well as the DIAR system, in terms of transactions, block time, tokens they use, and other important characteristics.

To provide practical transaction time data, we need specific information such as the *blocktimestamp* of when the transaction was mined and the details of the transaction. In Ethereum, one can get the *blocktimestamp* using *block.timestamp* within the solidity smart contract or by querying the blockchain explorer or node API. If we have a specific hash of a transaction, we can use a blockchain explorer like *Etherscan* to see the details of that transaction, including the timestamp when it was mined.

In Table 3 we have provided the details of the smart contract where the generation and verification of diplomas have been tested in the same file. We must emphasise the fact that depending

INPUT: Solidity Contract Code, Deployment Settings

OUTPUT: Gas Usage per Function

```

1.  CONFIGURE Hardhat (Solidity Version, Network,
    Accounts)
2.  INITIALIZE Ethers Library
3.  FUNCTION main():
4.      ATTEMPT Deploy Smart Contract -> Contract
        Address
5.      ELSE LOG ERROR "Deployment Failed"
6.      GET Deployed Contract Instance at Contract Address
7.      CALL TestGasUsage(Contract Instance)
8.  END FUNCTION
9.  FUNCTION TestGasUsage(Contract):
10.     GET Initial Gas Price
11.     GET Total Transactions
12.     DEPLOY Test Contract (if needed for specific tests) ->
        Test Contract Instance
13.     GET Initial Ether Balance (if applicable)
14.     // Loop through each function in the Smart Contract:
15.     FOR EACH Function IN Smart Contract:
16.         RECORD Start Gas
17.         ATTEMPT Call Function
18.         RECORD End Gas
19.         CALCULATE Gas Used = End Gas—Start Gas
20.         LOG "Function [FunctionName] Gas Used: ", Gas
            Used
21.     END FOR
22.     // Log final balance or other relevant metrics
23. END FUNCTION
24. EXECUTE main()

```

on the code, the elements we insert and the method of verifying the diplomas, the gas fee also changes, which depends above all on the execution of the code in the smart contract. The more code that needs to be executed, the more the costs of executing the smart contract increase.

The duration of the transaction depends on several factors, such as the performance of the computer on which the smart contract is executed, the performance of the network, the architecture of the system, and the security and cryptographic mechanisms [35]. The proposed framework provides a straightforward solution for HEIs to store and verify diplomas securely. It ensures that the data remains unchanged and can be easily checked for accuracy since its creation. The classification of data in the DIAR framework is off-chain and on-chain.

Deploying fully on-chain offers faster verification and stronger guarantees of immutability and decentralisation, but it comes at a significantly higher cost, making it impractical for higher

education institutions. In DIAR, we therefore separate on-chain and off-chain storage: only the hash and CID, essential for rapid verification, are recorded on-chain, while all additional data resides in off-chain IPFS. From an HEI management perspective, this hybrid approach is both financially sustainable today and expected to become even more cost-efficient in the future [36].

6 | Discussion

Through this research, we have attempted to provide a blockchain solution for the generation and verification of diplomas as important documents in HEIs, taking into account the various challenges and problems that blockchain technology is facing during practical implementations for the generation and verification of diplomas in HEIs. Very important in the DIAR framework, but also in blockchain systems in general, is the digital signature and the mechanisms used to protect the identity, privacy and security of personal data. The digital signature of diplomas, by the management, is carried out using ECDSA, with the sole purpose that in these cases it is much easier to apply and more adaptable when using the Ethereum platform, and above all, it enables the use of private or public keys during the hashing of the diploma, which is important in such blockchain systems, and especially in the DIAR framework during the generation and verification process [37]. The DIAR framework also uses cryptographic mechanisms to encrypt and decrypt data, whether on-chain or off-chain, providing even higher security through AES, SHA-256 and zero-knowledge proof or homomorphic encryption. The reason why we have used these algorithms for encryption is that AES enables secure encryption of diploma data while simultaneously providing secure transfer to the recipient, efficiency, and faster processing [38]. Meanwhile, SHA-256 plays an important role in maintaining data integrity, digital signatures and connecting the system to the blockchain network [28].

The main findings in this paper propose a practical solution that can be easily integrated into current systems. This solution involves storing diplomas as tokens and quickly verifying them using ID and hash values, reducing long waits and administrative work. Through DIAR, it can be easily and quickly verified, based on the diploma ID number and unique hash value, that the same diploma was generated by the respective HEI and that it belongs to the respective student. In comparison to other systems [3, 11–21], we believe that DIAR presents an easy solution, efficient and very integrable, that can be implemented to ensure the integrity and reliability of diploma data and at the same time, simpler and faster generation and verification.

All blockchain solutions have their own advantages and disadvantages in terms of generating and verifying diplomas. Diplomas are not cryptocurrencies, and executing such transactions is much more difficult and challenging, as errors in the data, whether intentional or unintentional, are difficult or almost impossible to correct with the same hash value [39, 40]. Table 4 presents a summary of the characteristics of the blockchain solutions that are included in our research, including DIAR, describing the advantages and disadvantages of each in terms of security, transparency and efficiency, identity and data access and even tokenisation, which is one of the main characteristics that we used for the DIAR system.

TABLE 2 | Comparison of blockchain platforms including the DIAR blockchain framework.

	Ethereum	Solana	Hyperledger fabric	DIAR framework
TPS (transactions per second)	100–1,000	2,000–3,000	500–3,000	6-7 diploma generation transactions per second
Max theoretical TPS	10,000+	65,000+	20,000+	2000-7000 Layer-2 TPS
Block time	12.06s	400 milliseconds	Configurable	12–15 s (block time); up to several minutes generation, Milliseconds to a few seconds verification
Time to finality	1–2 min (proof of work)	1 s	near-instant	12 s to 1 min
Governance model	off-chain	off-chain	permissioned, specified by users	on-chain and IPFS
Token	ETH	SOL	no native token	non-fungible token

TABLE 3 | Details from the smart contract implementation.

	Insert a certificate, verify by index number, verify by certificate number	Verification smart contract (modifiers, exporters, find document hash, get exporter info)
transaction hash	0x8b2256054696fd8d78d7fcf51961070cb15d3af87ba259b3521d6b2baca3ee6e	0x9f5c600a77975318c3c8d7c3727170a81bf7e9a866651371d87bcd76cb0d5b9
block hash	0x918ee0afd907fa0f2b056b5ac1d6c47ac0e8701b143e3ab9c2e559b584d494e7	0xd96544757fd03495ec8c357ed8a861adf098a7778d348ab1cc1bee50b280bf53
contract address	0xd9145CCE52D386f254917e481eB44e9943F39138	0xd8b934580fcE35a11B58C6D73aDeE468a2833fa8
from	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to	DiplomaRegistry.(constructor)	Verification.(constructor)
gas	644549 gas	1710808 gas
transaction cost	560477 gas	1487659 gas
execution cost	470903 gas	1332157 gas

TABLE 4 | Main characteristics of blockchain solutions including DIAR blockchain framework.

Study	Blockch. solution	Gener. diploma	Verif. diploma	Iden. and access	Sec.	Effic. and exec.	Dig. Sign.	Token.
[3]	UAEchain	×	×	✓	✓	✓	✓	×
[11]	Educert-chain	✓	✓	✓	✓	✓	✓	×
[12]	VECefblock	✓	✓	✓	✓	✓	×	×
[13]	DVD	×	✓	✓	✓	×	✓	×
[14]	BeCertify	✓	✓	✓	×	✓	✓	×
[15]	Digital Token	✓	✓	✓	✓	×	×	✓
[16]	Creducate	✓	✓	×	✓	✓	×	✓
[17]	QualiChain	✓	✓	✓	✓	✓	✓	×
[18]	IU-TransCert	✓	✓	✓	✓	✓	✓	×
[19]	Credere	✓	✓	✓	✓	✓	×	×
[20]	B4E	✓	✓	✓	✓	✓	✓	×
[21]	DocStone	✓	✓	✓	✓	✓	✓	×
	DIAR	✓	✓	✓	✓	✓	✓	✓

As can be seen, blockchain solutions for diploma generation and verification share several key features that make them attractive. At their core, these systems leverage cryptographic mechanisms to ensure data security, providing a robust foundation for protecting sensitive academic information. They also incorporate self-sovereign identity mechanisms, allowing individuals to maintain control over their personal data while still enabling authorised access. One of the properties of these blockchain-based systems is their use of digital signatures. These signatures, typically from key actors in the academic ecosystem, add an extra layer of authenticity and trust to the diplomas. The efficiency of these systems is also noteworthy, with many offering rapid execution of transactions and verifications. However, these solutions are not without their challenges.

The scalability of blockchain systems, particularly when dealing with large volumes of data over extended periods, remains a significant hurdle. Layer-2 solutions like Arbitrum, zkSync, and Polygon offer greater scalability, lower costs and faster diploma processing. However, their integration with DIAR introduces challenges such as complex data transfer between L1 and L2, off-chain data handling, and the need for standardisation and stable cryptographic protocols. This limitation stems partly from the inherent constraints of current blockchain platforms. Additionally, the practical implementation of blockchain for managing big data presents its own set of difficulties, potentially hampering the long-term sustainability of these solutions in the education sector.

When it comes to the implications, the same are more diverse; however, as far as technological implications are concerned, DIAR has an efficient performance in terms of the testing we have done in the Sepolia test network. How will DIAR's performance be in generating and verifying diplomas in large quantities, it remains a challenge for the future to be explored in terms of testing with particular HEIs. What is more important, with regard to interoperability with existing platforms, is that DIAR can easily be integrated into existing platforms as a subdomain of LMS (Learning Management System) and existing systems, but obviously, for its operation, hardware upgrades and additional tools need to be installed in HEI networks.

The practical implementation and evaluation of DIAR reveal clear benefits over existing blockchain credentialing systems. DIAR's capacity to encapsulate the complete diploma as a single, verifiable token allows for instant and secure verification, bypassing the delays inherent to models like VerDe [9], where each course credit must be validated individually. Compared to Educert-chain [11], BeCertify [14], and other solutions listed in Table 4, DIAR's architecture simplifies both integration and day-to-day institutional workflows by providing flexible user roles, automated digital signing, and comprehensive auditing tools. Additionally, by supporting advanced encryption and privacy measures (e.g., zero-knowledge proofs and one-way hash functions), DIAR mitigates data leakage and upholds the privacy of diploma holders, addressing gaps in existing platforms. Its adaptability as either a standalone module or a subdomain for institutional IT ecosystems further underscores its practical value. Thus, DIAR establishes itself as a future-ready solution, uniquely responsive to evolving educational practices and regulatory expectations, while maintaining operational efficiency and enhancing trust in digital credentials.

Regarding the economic and legal implications, it is obvious that the complete implementation of DIAR, as well as its integration and commissioning into existing systems, has much higher costs compared to centralised systems; however, for the moment, blockchain systems are seen as the only alternative for the complete digitalisation of diplomas [1, 41, 42]. Undoubtedly, the implementation of DIAR in any country must also be in accordance with the legal legislation of that country. Each country establishes its own regulations for diploma issuance and verification, which are then implemented by institutions in accordance with national standards and oversight bodies. Achieving greater standardisation of diploma formats and procedures across all higher education institutions would significantly simplify administration and verification, yet remains a major challenge in many regions.

DIAR is designed to align with the European Digital Credentials for Learning (EDC) framework, embracing the issuance of verifiable, tamper-proof diplomas that meet EU standards for digital credentials [43]. By adopting EDC data and JSON-LD signatures, DIAR ensures that diplomas are interoperable and easily verifiable by employers, institutions, and stakeholders across Europe. Furthermore, DIAR's implementation supports compliance with accreditation requirements and leverages accreditation databases, such as Qualification Dataset Register (QDR) and Database for External Quality Assurance Result (DEQAR), so that only degrees from externally validated, accredited programmes or institutions can be issued through DIAR [44]. This approach not only guarantees trust and transparency but also enables cross-border authentication and mobility, fully supporting the goals of the European Commission's digital transformation agenda for higher education.

The ethical implications of DIAR, in contrast to centralised systems, are significantly enhanced due to the decentralisation of data; yet, there is an increasing focus on the automation of services, which diminishes human involvement and access within these systems [45, 46].

6.1 | Future Work and Limitations

Unlike other practical solutions included in Table 1 and [1, 2], the DIAR system also has its own limitations in terms of cost, blockchain platform, lack of developers and users of blockchain systems, adaptation and transfer of data from current systems, and lack of database for management of larger quantities of diplomas. The DIAR system works on the Ethereum platform, a platform that is more open, more flexible, more transparent, secure and allows for easier integration into centralised systems.

Analysis, comparisons and the implementation of the DIAR reveal that the real use of blockchain is more complex than it seems in principle. The practical application faces several problems, namely the absence of a blockchain database capable of storing substantial volumes of data. IPFS is especially appropriate for situations where conventional centralised servers can struggle to manage substantial data volumes or when redundancy and availability are critical [47, 48]. In this regard, it may serve as a resolution to significant data difficulties. Nevertheless, it is not just designed for large data. IPFS serves other functions beyond

large data storage, including site hosting, content distribution and the development of decentralised apps. IPFS can manage large data sets; it is not its only function. This is a limitation of DIAR, but, in reality, none of the blockchain systems provide a superior alternative, except for a few that use centralised databases.

The second limitation of blockchain solutions is that they cannot update or save data and display it on demand [49, 50]. This is because practically all the data, before being stored in the form of tables, classes, each of them has a unique hash value that is generated in the network. Each unique hash value is linked to the corresponding hash value of the previously entered data, thus creating an interconnected network, which, unlike centralised systems, is much more difficult to manage, and access to the data is much more difficult. Even in the DIAR framework, a diploma, if stored as a token, is impossible to modify or delete.

Third, in terms of data security, blockchain systems are much safer because it is not possible to change the data and not to confirm it with other nodes in the network [51, 52]. Unlike traditional systems, which can be changed directly from the database, it is very difficult to distinguish such a change. So, any change to the data in the network must be confirmed by all nodes that are part of the network, and only then can the data be added, while practically the process of taking away diplomas is impossible, and it is generated again with the same hash value. Revoking of diplomas is only expressed by the regeneration of the diplomas, with another unique hash value.

DIAR provides a streamlined approach for generating and verifying diplomas using blockchain. Instead of direct revocation, DIAR handles updates by regenerating the diploma with a new hash value and making the previous version inaccessible; the old diploma remains on the blockchain but is no longer valid. To keep costs manageable for higher education institutions, only minimal data, hashes and content identifiers, are kept on-chain, while all supporting details are stored off-chain. This structure makes diploma verification fast and allows updates or invalidations without changing the system's architecture, as only off-chain metadata and hashes need to be refreshed. For students and verifiers, the process is seamless; they receive the new diploma's hash and need not interact with revoked records. Administrators update the relevant data and issue a new hash via smart contracts, ensuring invalid diplomas cannot be accessed or verified. While DIAR does not solve all practical blockchain challenges, it specifically addresses HEIs' difficulties with secure, efficient diploma generation and integration with legacy systems. DIAR can even function as a separate system specifically for this service, not integrating it at all into the LMS and other systems that HEIs possess. Moreover, DIAR can be implemented even only in the diploma generation offices, which would make the work of many administrators, but also the university management itself, easier. Through practical implementation, we have verified that the system is fully functional. Our next step is to conduct tests using specific case studies, where we will take real diplomas and evaluate the real-time generation and verification process for multiple documents.

This blockchain system has the potential to extend beyond higher education; it can be applied in fields such as medicine, finance and any area that requires robust data management and real-time

processing. By creating a nationwide blockchain platform, we could effectively manage all higher education institutions, both public and private. This would help address issues related to data misuse, fraudulent scientific papers and the improper awarding of academic titles, ultimately benefiting both students and faculty across various HEIs. As further work remains:

- The general implementation of DIAR, which requires a lot of teamwork and interaction with education institutions, high costs for practical implementation, and comprehensive implementation in LMS and current systems to be gradually replaced with blockchain systems.
- The implementation of various intelligent agents, in order to prevent various attacks, but above all to help the main actors in using the system, easier identification and creating easier and more attractive, yet more secure interfaces that guarantee privacy and protection from various attacks through intelligent algorithms.
- Performance analysis of the DIAR system is currently limited to testnet results. Future work will focus on practical implementation within a higher education institution, where detailed evaluations, including scalability stress tests, platform cost comparisons, gas usage for diploma operations and other key performance metrics, will be conducted under real conditions.
- Through the combination of edge intelligence, BT and AI techniques, security, transparency and efficiency can be increased, but above all, low-latency and critical-computation can be affected. Above all, in this direction, the scalability of the DIAR system can be affected, because in addition to the system being able to effectively manage the quantity of academic credentials generated or verified, it can also guarantee the capacity to handle large volumes of data.
- A usability survey and structured user feedback are planned in future DIAR deployments to systematically evaluate user experience and identify potential improvements for both administrators and students. In addition, Nielsen's usability heuristics will be applied to rigorously assess interface design and functionality [53].

In summary, the successful comprehensive scaling of DIAR necessitates navigating a fundamental philosophical challenge inherent in decentralised ledger technologies: the enduring tension between immutable integrity and the pragmatic management of substantial data volumes, a hurdle amplified by the current absence of a dedicated, high-capacity blockchain database. The current architecture addresses this by maximising efficiency, utilising IPFS for decentralised storage and minimising transaction costs by tokenising the diploma as a single NFT. Our next task remains to determine whether DIAR will be integrated within the LMS as a plugin or widget, allowing students and administrators to interact through a simple interface, or function as a standalone system that can be connected as a subdomain to existing infrastructure. For future large-scale growth, DIAR is planned to integrate edge intelligence, BT and AI techniques to enhance efficiency and manage big data volume, aiming toward high Layer-2 throughput. To further improve scalability and reduce costs, future work will explore integration of Layer-2 solutions. Approaches such as Arbitrum, zkSync and Polygon

offer greater efficiency, but implementation will require addressing data transfer, standardisation and integration complexity between on-chain and off-chain components. In the short term, the system can be deployed in phases, either as a separate diploma service or within diploma generation offices at HEIs. The immediate next step involves conducting real-time tests using specific case studies with multiple real diplomas. Ultimately, achieving robust national scalability depends on both ongoing technological advancement and resolving policy heterogeneity through the standardisation of diplomas across higher education institutions, simplifying the legal and operational compliance necessary for widespread, consistent deployment.

7 | Conclusion

The creation of blockchain systems for managing big data remains a significant challenge in education, particularly for higher education institutions. Diplomas are high-value documents and are frequently targeted for abuse and fraud. Only blockchain technology, with its attributes of immutability, traceability and consensus, can robustly safeguard diplomas and related data from misuse within and outside institutions.

This study provides a simplified, cost-effective approach for diploma administration using blockchain, acknowledging both the advantages and present limitations of such systems. While obstacles to wide-scale deployment persist, the DIAR system streamlines diploma generation and verification for HEIs, leveraging the essential strengths of blockchain while minimising implementation difficulty and cost.

Our work documents the full technical process, including practical architecture and code sequences, which are often missing in similar research. Actual implementation steps and tested scenarios offer clear guidance to future developers and HEI managers seeking modern credential solutions. A central innovation of DIAR is the tokenisation of the entire diploma as a single, indivisible digital asset on the blockchain via smart contracts and IPFS. This whole-diploma approach significantly simplifies integration with existing data management systems, reduces operational steps, and allows instant, tamper-proof verification for all stakeholders. With this design, DIAR guarantees strong immutability, integrity and secure diploma issuance, while offering rapid, fraud-resistant verification that sets it apart from other methods.

DIAR's practical design demonstrates a future-ready path for digitising diplomas and may extend to other fields needing secure digital documents. However, comprehensive performance testing in live HEI environments, shaped by regional regulations and ethics, remains a priority for future research. With continued refinement, DIAR has the potential to transform academic credential management and support greater trust, transparency and efficiency in higher education.

Author Contributions

Avni Rustemi: conceptualisation, writing – original draft, reviewing and editing. **Fisnik Dalipi:** conceptualisation, reviewing and editing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data generated or analysed during this study are included in this article.

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